

Project Integration Submission Packet

Smart Pothole Monitoring System Team

May 22, 2026

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README.md File

Source: ../README.md

Pothole Detection and Reporting Platform

Team Information

Name	Email	Role
Atem Ako	member1@school.edu	Project Manager
Amir Abdalla	member2@school.edu	Backend Developer
Nahaeli Brunder	member3@school.edu	Systems "Engineer"
Sthapanavichet Long	member4@school.edu	QA and Documentation

Project Description

Road damage is a daily safety and cost issue for drivers, cyclists, delivery workers, and municipalities.

This project provides a simple platform that helps people report potholes with clear location details, photos, and descriptions.

At a high level, the platform acts as a bridge between identification and the maintenance teams. Teams can use the platform to track and manage reported potholes.

Working Agreement

Source: ../docs/WORKING_AGREEMENT.md

Working Agreement

Meeting Information

- Weekly meeting day/time: Tuesdays at 7:00 PM (America/New_York)
- Meeting platform: Discord voice channel
- Backup platform: Microsoft Teams

Communication Expectations

- Preferred channels: Discord for daily communication, GitHub comments for issue/PR context
- Expected response time: within 12 hours on weekdays, within 24 hours on weekends
- Urgent blockers: tag '@here' in Discord and open a GitHub issue immediately

Accountability Rules

- Missing meetings without notice: responsible for publishing next meeting notes and action summary
- Repeated lateness (3+ times): temporary reduction in planning vote priority for that sprint
- Lack of participation: task reassignment plus documented recovery plan
- Failure to respond to teammates: escalation to team lead, then professor if unresolved

GitHub Workflow Agreement

- No direct pushes to 'main'; all changes must go through Pull Requests
- Minimum 1 reviewer approval required before merge
- PR must include linked issue and checklist completion
- Branch naming convention:
 - 'hardware/<short-description>' for CAD or electronics integration work
 - 'firmware/<short-description>' for embedded code updates
 - 'software/<short-description>' for dashboard/backend/frontend updates
 - 'docs/<short-description>' for documentation updates

Testing Requirements

- Before merge, contributor must verify acceptance criteria from linked issue
- Hardware changes:
 - CAD fit check completed
 - IoT module placement and mounting constraints verified
- Firmware changes:
 - Build/flash verification completed
 - Sensor read + telemetry transmission validated on test device
- Software changes:
 - Relevant unit/integration tests pass
 - Manual smoke test of impacted dashboard flow completed
- Reviewer is responsible for confirming testing evidence is included in PR

Current Changelog

Source: ../docs/CHANGELOG.md

Changelog

All notable changes to this project will be documented in this file.

2026-05-21

Added

- Created initial project documentation structure in local repository.
- Added 'README.md' with team table and non-technical project overview.
- Added working agreement at 'docs/WORKING_AGREEMENT.md'.
- Added project backlog templates and issue drafts for GitHub import.
- Added planning artifacts for milestones and Kanban board setup.

Links

- Pull Request: [PR #1](https://github.com/your-org/your-repo/pull/1)

Issue Screenshots

One screenshot of a fully written issue per student. Each screenshot should clearly show issue title, issue body, and assigned milestone (if applicable).

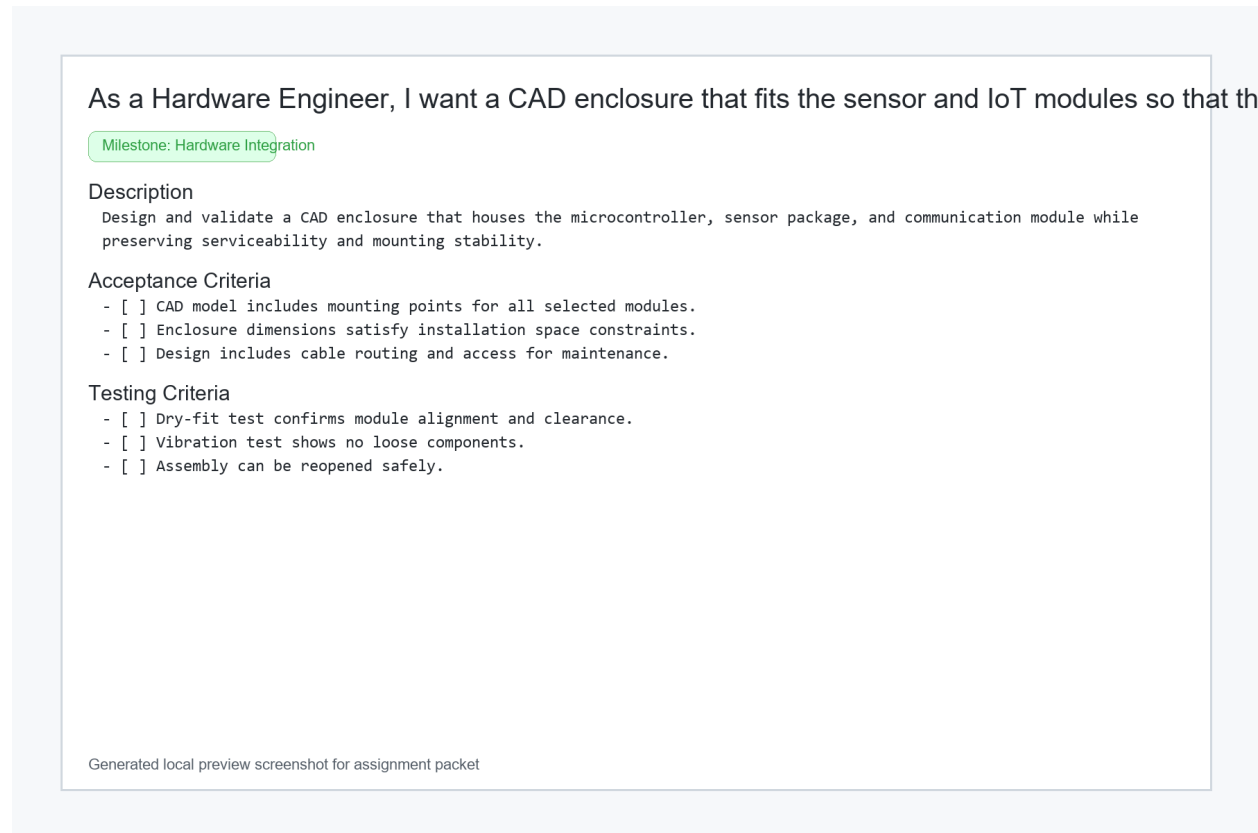


Figure 1: Student 1 Fully Written Issue

As a Firmware Developer, I want embedded firmware to collect accelerometer and GPS data so

Milestone: Firmware Data Pipeline

Description

Implement firmware tasks for sampling motion/location data, deriving pothole events from thresholds, and transmitting event payloads through IoT connectivity.

Acceptance Criteria

- ☐ Firmware reads accelerometer and GPS at intervals.
- ☐ Threshold logic triggers pothole events.
- ☐ Payload includes timestamp, location, severity.

Testing Criteria

- ☐ Firmware builds and flashes to target board.
- ☐ Bench vibration triggers event creation.
- ☐ Payload is received by ingestion endpoint.

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Figure 2: Student 2 Fully Written Issue

As a Data Reviewer, I want a dashboard that visualizes pothole events on a map and timeline so

Milestone: Software Visualization

Description

Create a software dashboard that displays ingested events with map markers, severity coloring, and timeline filtering.

Acceptance Criteria

- ☐ Dashboard lists location, timestamp, severity.
- ☐ Map view renders severity pins.
- ☐ Timeline filter updates results correctly.

Testing Criteria

- ☐ UI works with sample and live data.
- ☐ Filter counts stay consistent.
- ☐ Responsive on desktop/mobile widths.

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Figure 3: Student 3 Fully Written Issue

As a Project Integrator, I want a validated hardware-firmware-software data pipeline so that sens

Milestone: End-to-End Validation

Description

Run end-to-end integration tests from sensor trigger to dashboard display and document failure points with corrective actions.

Acceptance Criteria

- ☐ Generated field event appears within target latency.
- ☐ Failed transmissions are retried.
- ☐ Integration test steps are documented.

Testing Criteria

- ☐ At least 5 end-to-end runs recorded.
- ☐ One network interruption test completed.
- ☐ Evidence attached to issue or PR.

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Figure 4: Student 4 Fully Written Issue

Milestones Screenshot

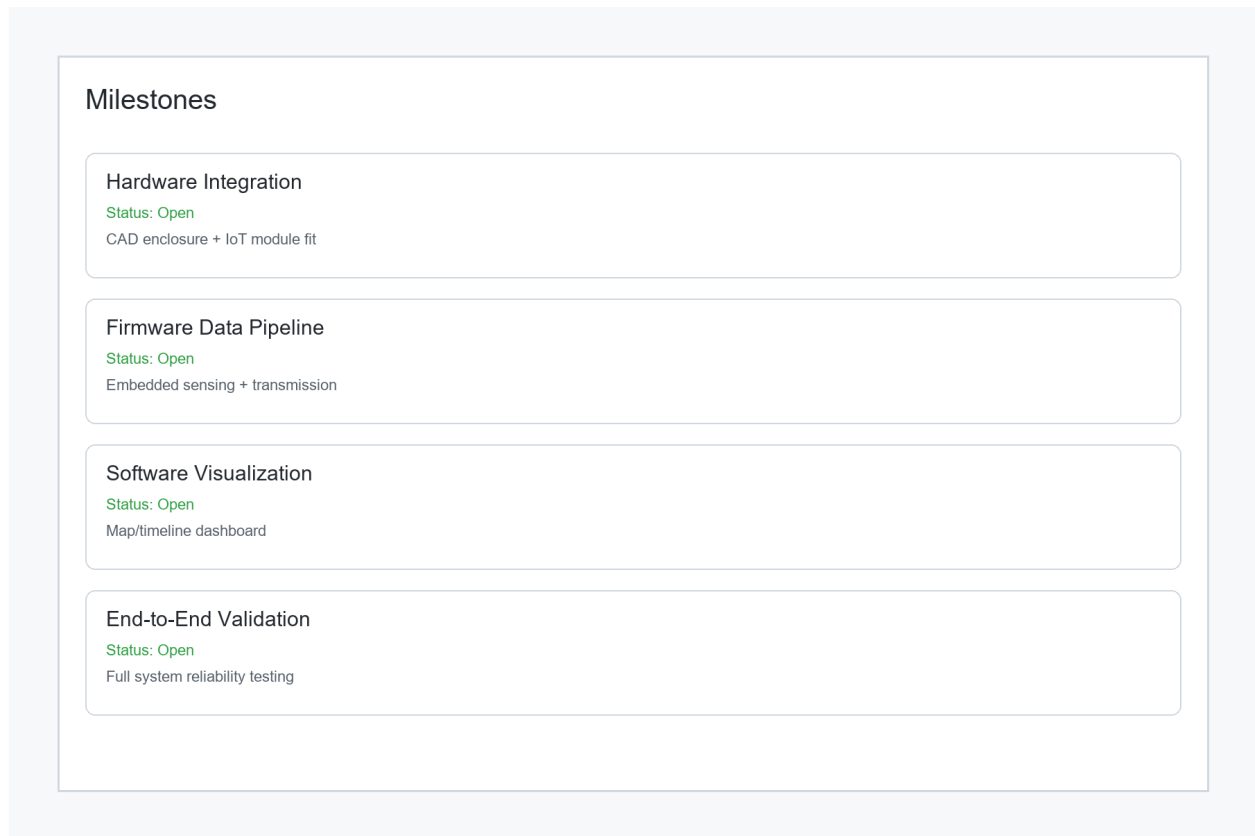


Figure 5: GitHub Milestones View

Kanban Board Screenshot

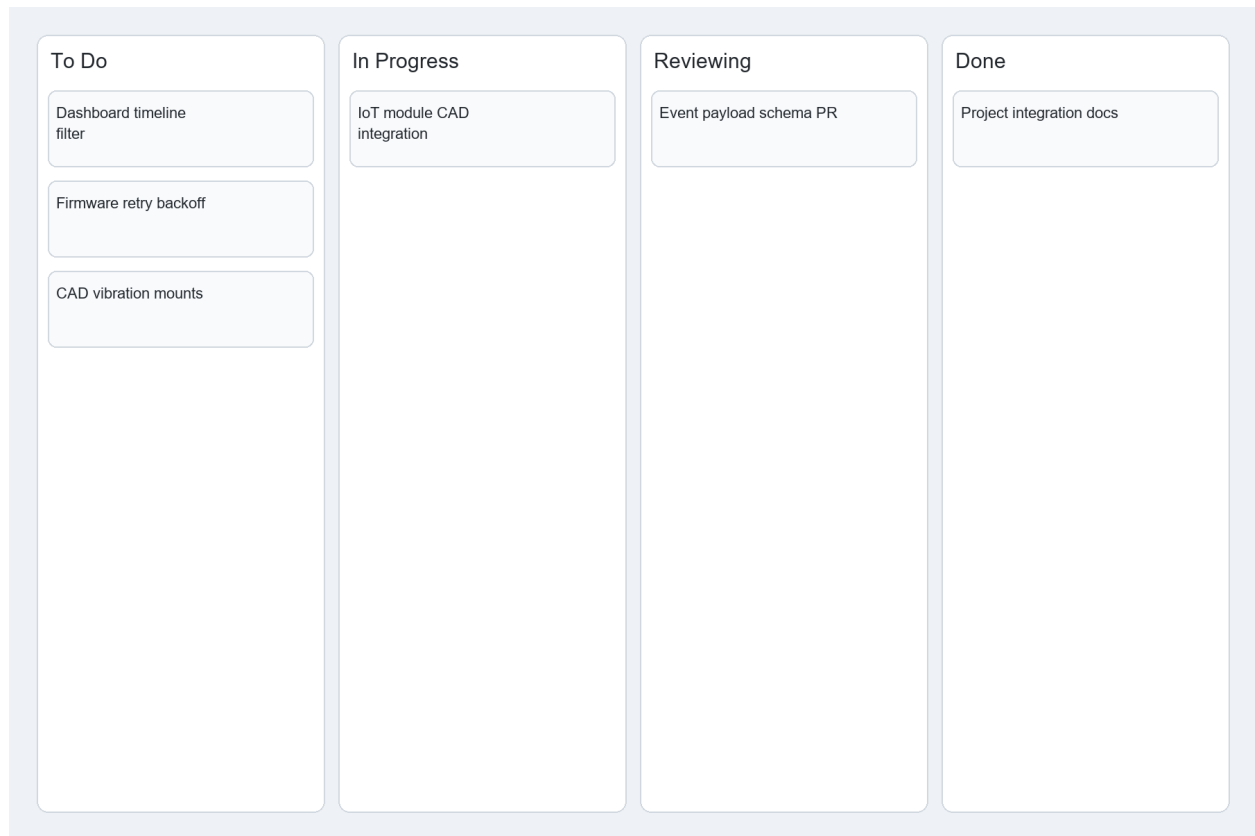


Figure 6: GitHub Kanban Board View